

An Autonomous Security Agent for SOC and Data Center Threat Intelligence and Response

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Abstract—Security Operations Centers (SOCs) and data centers face an increasingly unfavorable operational equation: high-volume heterogeneous telemetry, sophisticated adversaries blending into baseline activity, and response workflows that still depend heavily on human triage. This report proposes an Autonomous Security Agent (ASA) that integrates telemetry ingestion, threat intelligence fusion, machine learning and deep learning based detection, evidence correlation, explainable decision-making, and automated response orchestration. The design leverages streaming data infrastructure for scalable ingestion, STIX/TAXII for interoperable cyber threat intelligence exchange, and operational threat intelligence platforms such as MISP and OpenCTI. To improve analyst trust and governance, explainability is treated as a first-class requirement through SHAP and LIME style interpretation. The proposed architecture follows a closed-loop “sense–decide–act–learn” model and aligns incident response with modern cybersecurity governance and risk management guidance. This report presents the problem background, research objectives, solution architecture, implementation plan, expected outcomes, and evaluation methodology for a practical and reproducible autonomous cyber defense system suitable for SOC and data center environments.

Index Terms—Autonomous Security Agent, SOC, Data Center Security, Threat Intelligence, Incident Response Automation, STIX/TAXII, MISP, OpenCTI, Kafka, Explainable AI, SHAP, LIME, APT Detection

I. INTRODUCTION AND BACKGROUND

A. Problem Description

Modern cybersecurity operations have evolved into a large-scale data engineering and decision-making problem. Security teams must process network logs, host events, identity telemetry, cloud signals, and external threat intelligence feeds in near real time. However, many SOC environments still rely heavily on human-driven alert triage and manual playbooks, resulting in delayed detection, alert fatigue, and inconsistent incident handling.

Recent studies emphasize that cybersecurity must move from static rule-centric defense toward data-driven and automated intelligence workflows. Sarker [1] describes cybersecurity data science as a discipline that combines large-scale security data collection, preparation, modeling, and adaptive learning to support intelligent decision-making. Sarker later further argues that machine learning should support not only detection, but also prediction, prevention, and incident response automation [2].

Threat intelligence has emerged as a key mechanism for improving cyber defense by integrating disjointed cyber data

into evidence-based insights. Ibrahim *et al.* [3] explain that threat intelligence helps answer questions such as who the adversary is, what assets are at risk, how compromise occurs, and how defenders should detect or respond. However, they also identify major limitations: cyber data complexity, poor data quality, scalability issues, and the risk that low-quality intelligence will increase false alarms instead of reducing them.

In SOC and data center practice, the operational gap is clear. Existing systems often detect isolated alerts but fail to correlate them into coherent attack narratives or transform them into safe, explainable, and auditable response actions. This becomes especially critical in the presence of advanced persistent threats (APTs), botnets, and stealthy multi-stage intrusions, where single-event signatures are insufficient.

B. Research Gap

Current research and industrial practice contain several gaps:

- Many works focus only on intrusion detection, without integrating detection with automated incident response.
- Threat intelligence is often treated as an external feed rather than as a deeply integrated reasoning component.
- Existing SOC automation systems usually depend on static playbooks and lack adaptive learning.
- Explainability is still under-emphasized, despite being necessary for analyst trust and governance.
- Few practical system designs combine streaming ingestion, ML/DL-based detection, TI fusion, graph-based reasoning, and safe response orchestration in one end-to-end architecture.

Therefore, there is a need for a complete software and research design for an autonomous cyber defense agent that can ingest telemetry, fuse intelligence, detect threats, reason over evidence, and trigger constrained response actions in SOC and data center environments.

C. Research Objectives

The main objective of this project is to design an Autonomous Security Agent (ASA) for SOC and data center threat intelligence and response.

The specific objectives are:

- To design a scalable architecture that ingests and normalizes multi-source security telemetry in near real time.

- To integrate operational and standards-based threat intelligence into a unified machine-consumable representation.
- To build a hybrid detection stack combining anomaly detection, supervised classification, and graph-based reasoning.
- To design a decision engine that maps evidence to safe and auditable response actions.
- To incorporate explainability mechanisms that make model outputs understandable to SOC analysts.
- To define an evaluation framework using public cybersecurity datasets and SOC-relevant operational metrics.

D. Research Questions

This project is guided by the following research questions:

- 1) What existing methods can be used to detect threats, fuse intelligence, and automate incident response, and what are their strengths and weaknesses?
- 2) What architecture and technologies should be proposed for an autonomous security agent in SOC and data center environments?
- 3) How can the proposed solution be implemented in a practical and reproducible way?
- 4) How should the effectiveness of the solution be evaluated in terms of both machine learning quality and SOC operational outcomes?

E. Scope of the Project

This project focuses on the design and prototype-level implementation plan of an autonomous security agent for SOC and data center use cases.

The project includes:

- Telemetry ingestion and normalization
- Threat intelligence fusion
- Threat detection and classification models
- Correlation and reasoning over alerts and events
- Explainable decision support
- Automated response orchestration with safety guardrails
- Evaluation design using public datasets

The project does not include:

- Full production deployment across a real enterprise SOC
- Proprietary vendor integrations requiring closed commercial APIs
- Large-scale live response execution in operational critical infrastructure
- Formal red-team/blue-team exercises in a real environment

II. LITERATURE REVIEW AND PROPOSED SOLUTION

A. Methods to Solve the Problem

Several categories of methods have been used in cybersecurity research and practice.

1) *Rule-Based and Signature-Based Detection*: Traditional IDS/IPS systems such as Suricata provide deterministic detection for known threats through signatures and rules [17]. These systems are fast and interpretable but perform poorly against unknown or stealthy threats.

Advantages:

- High precision on known patterns
- Deterministic and explainable
- Operationally mature

Disadvantages:

- Cannot generalize well to unseen attacks
- Limited contextual reasoning
- High maintenance cost for rule updates

2) *Machine Learning and Deep Learning Detection*: Machine learning supports anomaly detection, attack classification, and predictive analytics [1], [2]. Unsupervised models can detect suspicious deviations, while supervised models can classify known attacks.

Advantages:

- Better ability to detect unseen patterns
- Useful for large-scale telemetry
- Can support adaptive learning

Disadvantages:

- Sensitive to data quality
- May produce false positives
- Often difficult to explain

3) *Threat Intelligence Driven Detection*: Threat intelligence helps enrich raw telemetry with adversary context, observables, malware families, techniques, and relationships [3]. Standards such as STIX/TAXII allow machine-readable exchange, while MISP and OpenCTI support operational intelligence management [10]–[12].

Advantages:

- Improves context and prioritization
- Supports correlation across multiple signals
- Useful for sharing and reuse

Disadvantages:

- Feed quality varies
- Can introduce noise and false positives
- Requires normalization and confidence scoring

4) *Graph and Provenance Based Reasoning*: APT detection often requires reconstructing causal or temporal relationships among events rather than analyzing individual alerts in isolation. Provenance and graph-based approaches address this by linking process, file, network, and user actions [20], [21].

Advantages:

- Stronger support for multi-stage attack detection
- Better for narrative reconstruction
- More suitable for stealthy attacks

Disadvantages:

- High computational cost
- Complex graph construction and storage
- Harder deployment in large-scale environments

5) *Explainable AI*: Explainability techniques such as SHAP and LIME help analysts understand why the system made a prediction [18], [19]. This is especially important in SOC workflows where high-impact actions require trust and accountability.

Advantages:

- Improves analyst trust
- Supports governance and auditing
- Helps reduce triage ambiguity

Disadvantages:

- Adds computational overhead
- Some explanations may be unstable
- Does not fully eliminate model complexity

B. Proposed Method

Based on the limitations of existing approaches, this project proposes an **Autonomous Security Agent (ASA)** that combines the strengths of multiple methods in one architecture.

The proposed solution uses:

- Streaming telemetry ingestion for scalable data collection
- Threat intelligence fusion through STIX/TAXII, MISP, and OpenCTI
- A hybrid detection stack including anomaly detection, supervised classification, and graph-based reasoning
- Explainable AI for interpretable alert scoring and decision support
- Response orchestration with policy-based safety guardrails

C. Solution Description

The ASA is designed as a closed-loop cyber defense system consisting of four major phases:

- 1) **Sense**: Collect telemetry from network, host, identity, application, and external intelligence sources.
- 2) **Decide**: Detect, classify, correlate, and infer threat severity and response options.
- 3) **Act**: Execute safe response playbooks or create analyst cases for approval.
- 4) **Learn**: Use analyst feedback, drift signals, and new intelligence to update models and confidence scores.

This design directly addresses alert overload, fragmented context, delayed incident handling, and limited explainability.

III. SOFTWARE ARCHITECTURE

A. General Architecture

The proposed software architecture follows a modular event-driven design. It can be deployed as a microservices-based platform or as loosely coupled services connected through a streaming bus.

High-Level Architecture of the Autonomous Security Agent

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Telemetry Sources + Threat Intelligence Feeds → Kafka
Ingestion → Normalization/Enrichment →
Detection Models + Classification Models + Graph
Reasoning → Decision Engine →
Response Orchestrator + Case Management → Feedback
and Continuous Learning

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Fig. 1. Conceptual architecture of the proposed ASA.

B. Main Components

The main components of the system are:

1) *Telemetry Ingestion Layer*: This component collects data from Zeek, Suricata, host logs, authentication logs, application logs, and cloud logs. Apache Kafka is used as the stream bus for scalability and replay capability [15].

2) *Normalization and Enrichment Layer*: Incoming events are parsed, timestamp-aligned, schema-normalized, and enriched with asset context, user context, geolocation, and threat intelligence. This layer ensures that heterogeneous data can be processed uniformly.

3) *Threat Intelligence Fusion Layer*: Threat intelligence from STIX/TAXII feeds, MISP, and OpenCTI is transformed into structured entities and relationships. Confidence scoring, deduplication, freshness checking, and entity resolution are performed here [10]–[12].

4) *Detection and Classification Layer*: This layer contains:

- Anomaly detection models for unknown or suspicious behavior
- Supervised classifiers for known attack families
- Signature-based detection outputs from Suricata

5) *Reasoning and Correlation Layer*: This component links related events into attack hypotheses or narratives. Provenance graphs and ATT&CK-based behavior mapping are used to understand multi-stage intrusion progression [13], [20].

6) *Decision Engine*: The decision engine consumes model scores, graph context, asset criticality, and policy constraints to decide whether to enrich, alert, contain, or escalate.

7) *Response Orchestrator*: The orchestrator executes playbooks such as blocking domains, isolating endpoints, revoking credentials, or creating cases. Actions are constrained by risk policies and approval rules.

8) *Continuous Learning Module*: This module captures analyst feedback, tracks drift, and supports retraining and recalibration of the models.

C. Roles and Interactions

The ASA components interact as follows:

- 1) Sensors and feeds publish data into Kafka topics.
- 2) The normalization layer parses and enriches the data.
- 3) Features and graph relations are generated.
- 4) Detection and classification models score incoming events.
- 5) The reasoning layer aggregates evidence into contextual threat hypotheses.

- 6) The decision engine selects safe actions.
- 7) The response orchestrator performs actions or sends cases to analysts.
- 8) Analyst feedback is stored and later used for model improvement.

IV. TECHNOLOGY AND TOOLS USED

A. Programming Languages and Frameworks

- **Python:** Primary language for machine learning, data processing, and orchestration logic
- **PyTorch / TensorFlow:** Deep learning implementation
- **Scikit-learn:** Classical machine learning algorithms and preprocessing

B. Threat Intelligence and Security Platforms

- **MISP:** Threat intelligence sharing, IOC storage, event correlation [11]
- **OpenCTI:** Knowledge graph style CTI management based on STIX2 [12]
- **STIX/TAXII:** Intelligence representation and exchange standards [10]
- **MITRE ATT&CK:** Adversary behavior mapping [13]
- **MITRE D3FEND:** Defensive action mapping [14]

C. Data Collection and Streaming

- **Apache Kafka:** Stream ingestion, partitioning, replay, and decoupled processing [15]
- **Zeek:** Network security monitoring logs [16]
- **Suricata:** IDS/IPS and signature-based detection [17]

D. Explainability and Model Trust

- **SHAP:** Additive feature attribution for interpretable predictions [18]
- **LIME:** Local surrogate explanations for heterogeneous models [19]

E. Case Management and Response

- **TheHive or equivalent case management platform:** Alert-to-case workflow support
- **SOAR-style playbooks:** Safe automated response execution

F. Rationale for Technology Selection

These technologies are selected because they align with the system requirements:

- Kafka supports scalable low-latency event pipelines.
- MISP and OpenCTI provide mature open CTI ecosystems.
- STIX/TAXII ensures interoperability.
- Zeek and Suricata are widely used and compatible with SOC workflows.
- Python and major ML libraries enable rapid model development.
- SHAP and LIME support explainability required for analyst trust.

V. IMPLEMENTATION PLAN

A. Main Development Stages

The implementation plan is divided into five stages.

1) Stage 1: Requirements Analysis and Threat Modeling:

- Define SOC and data center use cases
- Identify telemetry sources and threat intelligence inputs
- Define attack scenarios and policy guardrails

2) Stage 2: Data Pipeline and Intelligence Integration:

- Set up Kafka-based ingestion
- Build normalization and enrichment pipelines
- Integrate STIX/TAXII, MISP, and OpenCTI connectors

3) Stage 3: Detection and Reasoning Development:

- Train anomaly detection and classification models
- Build graph/provenance based correlation module
- Integrate ATT&CK and D3FEND mappings

4) Stage 4: Response Automation and Explainability:

- Design safe response playbooks
- Implement decision engine
- Add SHAP and LIME based explanation outputs

5) Stage 5: Evaluation and Reporting:

- Conduct experiments on selected datasets
- Measure ML and operational metrics
- Finalize documentation and report

B. Expected Schedule (Preliminary)

TABLE I
PRELIMINARY PROJECT SCHEDULE

Milestone	Start	End	Duration
Requirements and threat model	Week 1	Week 2	2 weeks
Schema and normalization design	Week 3	Week 5	3 weeks
Kafka ingestion and TI connectors	Week 6	Week 9	4 weeks
Detection and classifier base-line	Week 10	Week 14	5 weeks
Graph reasoning module	Week 15	Week 18	4 weeks
Explainability and decision engine	Week 19	Week 21	3 weeks
Response orchestration integration	Week 22	Week 24	3 weeks
Experimentation and evaluation	Week 25	Week 28	4 weeks
Final report and refinement	Week 29	Week 30	2 weeks

C. Feasibility Assessment

1) *Technical Challenges:* Potential technical challenges include:

- Large-scale heterogeneous telemetry processing
- Data quality and label noise
- High false positive rates in anomaly detection
- Scalability of graph/provenance reasoning
- Safe automation without service disruption

2) Workarounds:

- Use Kafka partitioning and modular consumers for scalability
- Apply data validation and feature selection
- Combine anomaly detection with TI enrichment and rule guardrails
- Restrict high-cost reasoning to suspicious subsets
- Use autonomy tiers and human approval for high-impact actions

3) Resources Required:

- Hardware: workstation or server with adequate CPU/GPU and storage
- Software: Python, ML libraries, Kafka, MISP, OpenCTI, Zeek, Suricata
- Data: public cybersecurity datasets and synthetic test data

The required resources are accessible for research and prototype development, making the project feasible at academic capstone scale.

VI. EXPECTED RESULTS

Upon completion, the project is expected to deliver:

- A complete architectural design of an Autonomous Security Agent
- A prototype workflow for telemetry ingestion, enrichment, and intelligence fusion
- A hybrid detection framework combining anomaly detection, classification, and reasoning
- Explainable model outputs for SOC analyst interpretation
- A response orchestration design with policy guardrails
- An evaluation framework using realistic public datasets and SOC metrics
- A final capstone report in IEEE-style academic format

The final deliverables may include:

- Design document and software architecture diagrams
- Prototype scripts or modules
- Dataset preprocessing pipeline
- Experimental results and metric tables
- Final written report

VII. EVALUATION PLAN

A. Evaluation Goals

The solution will be evaluated against the research objectives in both technical and operational dimensions.

B. Datasets

The evaluation may use public datasets such as:

- UNSW-NB15
- CICIDS2017
- CSE-CIC-IDS2018
- CTU-13
- Bot-IoT
- LANL enterprise event data
- DARPA OpTC

C. Evaluation Criteria

1) Detection and Classification Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- False Positive Rate

2) Operational SOC Metrics:

- Time-to-Detect (TTD)
- Mean Time to Acknowledge (MTTA)
- Time-to-Respond / Contain
- Alert volume reduction
- Number of unsafe actions blocked by policy guardrails

3) System Metrics:

- Event throughput
- End-to-end latency
- CPU, memory, and storage cost

D. Validation Approach

The validation strategy includes:

- Baseline comparison with rule-based IDS and classical ML models
- Ablation study of TI fusion, graph reasoning, and explainability
- Practical case studies using multi-stage attack scenarios
- Analysis of explainability usefulness for analyst interpretation

VIII. DISCUSSION

The proposed ASA design provides a realistic direction for SOC modernization because it treats intelligence, ML, reasoning, and response as one integrated system rather than separate tools. The strongest value of the design lies in combining multiple strengths: real-time ingestion, contextual intelligence, adaptive detection, safe automation, and explainability.

However, the design also introduces trade-offs:

- Higher autonomy increases the risk of operational disruption.
- Stronger reasoning improves context but adds computational cost.
- Richer threat intelligence improves prioritization but also introduces noise and poisoning risks.
- More advanced models may improve detection but reduce interpretability.

Therefore, the ASA should not be treated as a fully independent black-box system. It should instead be built as a governed autonomous platform where automation is graduated by risk and where analysts remain in control of high-impact decisions.

IX. CONCLUSION

This report presented a complete capstone-style proposal for an Autonomous Security Agent for SOC and data center threat intelligence and response. The design addresses the limitations

of analyst-centric and rule-only security operations by combining telemetry ingestion, threat intelligence fusion, machine learning based detection, graph-based reasoning, explainable AI, and automated response orchestration.

The proposed architecture is aligned with modern cybersecurity data science principles, operational threat intelligence practice, and incident response governance. It is also structured to be academically rigorous and practically implementable. The project therefore provides a strong foundation for research and prototype development toward intelligent, explainable, and safer cyber defense automation.

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